# Ethan J. Moger

Queensbury, NY 12804 | (518) 955-1935 | ejm349@cornell.edu | www.linkedin.com/in/ethan-moger | Portfolio

## EDUCATION

**Cornell University | Duffield College of Engineering** — **Ithaca, NY**
*Major: Mechanical Engineering* — *Aug 2024 – May 2028*
Cumulative GPA: 4.264
Relevant Coursework: Dynamics, Statics, Thermodynamics, Differential Eqs, Linear Algebra, Physics I/II, Calculus I/II/III

**Queensbury Senior High School** — **Queensbury, NY**
*NYS Advanced Regents Diploma with Honors, Seal of Distinction in Math & Science* — *Sep 2020 – Jun 2024*
International Baccalaureate Diploma
Class Rank: 1 / 241 | Cumulative GPA: 99.320 / 100.000

## WORK & RELEVANT EXPERIENCE

**Fluor Marine Propulsion (Naval Nuclear Laboratory)** — **Niskayuna, NY**
*Incoming Mechanical Engineering Intern – Nuclear Reactor Equipment Design* — *May 2027 – Aug 2027*

**X-Ray Optical Systems** — **East Greenbush, NY**
*Engineering Intern* — *May 2026 – Aug 2026*

- Redesigned a replacement for an obsolete part in SolidWorks, reducing cost by 18% while meeting all design constraints
- Updated SolidWorks/PDM assembly files across 7 product lines, improving inventory management and documentation
- Verified regulatory compliance for ~900 parts and subassemblies, organizing records using Epicor and Excel

**Cornell University Hyperloop Project Team** — **Ithaca, NY**
*Magnetics Subteam Lead, Mechanical Engineer* — *Sep 2025 – Present*

- Directed magnetics development and coordinated cross-functional subsystem design, integration, and implementation
- Utilized Ansys Maxwell to design and optimize a linear induction motor, doubling thrust output versus prior designs
- Simplified Maxwell simulations by redesigning coil models and utilizing stranded conductors, reducing mesh complexity

**Mechanical Design Project: Spotted Lanternfly Control** — **Ithaca, NY**
*Cornell University MAE 2250 Student* — *Jan 2026 – May 2026*

- Designed and prototyped a handheld device to remove spotted lanternfly egg masses, with a capacity of 100 egg masses
- Developed CAD models in Fusion 360 and fabricated components through 3D printing, machining, and assembly
- Conducted testing and analysis, achieving 2.65s egg mass removal time and earning an Outstanding Design nomination

**Glove Dryer Design & Provisional Patent** — **Queensbury, NY**
*Member/Contributor* — *Sep 2022 – Jun 2023*

- Developed a soccer goalie glove-drying system through research and design with a focus on material preservation
- Designed and 3D printed the product in OnShape, incorporating feedback and advice from engineering professionals
- Presented the prototype to a panel of engineers and later applied for and was granted a provisional patent

## LEADERSHIP EXPERIENCE

**Physics 1112 / Physics 2213 Learning Assistant** — **Ithaca, NY**
*Undergraduate TA* — *Aug 2025 – May 2026*

- Facilitated collaborative problem-solving in discussion sessions to support peer learning in calculus-based physics
- Enhanced communication and teaching skills through weekly seminars on physics education theory and reflective practice
- Partnered with faculty to promote inclusive instruction and provide feedback to improve course delivery and engagement

**Adirondack Solar Alliance** — **Queensbury, NY**
*Co-Founder* — *Dec 2021 – May 2023*

- Co-founded a not-for-profit organization that has raised over $100,000 to install solar panels in Adjuntas, Puerto Rico
- Assembled and gave formal presentations to local businesses, securing financial support for past and future projects
- Developed the donation page, coordinated technology-related tasks for fundraising, and participated in the first installation

## SKILLS, ACTIVITIES, AWARDS & INTERESTS

**Skills:** SolidWorks, Ansys Maxwell, Fusion 360, MATLAB, OnShape, Python, Microsoft Office, Collaboration, Problem Solving
**Activities:** American Society of Mechanical Engineers, Intramural Soccer (Captain), Doubles Pickleball League
**Awards:** President's Award for Educational Excellence, RIT Science and Math Scholarship, AP Scholar
**Interests:** 3D Printing/Design, Programming, Mountain Biking, Skiing, Pick-Up Soccer, Pickleball, Cliff Jumping, Piano